

Single-column purification of free recombinant proteins using a self-cleavable affinity tag derived from a protein splicing element

Shaorong Chong ^a, Fana B. Mersha ^a, Donald G. Comb ^a, Melissa E. Scott ^a, David Landry ^a,
Luis M. Vence ^a, Francine B. Perler ^a, Jack Benner ^a, Rebecca B. Kucera ^a,
Christine A. Hirvonen ^{a,1}, John J. Pelletier ^a, Henry Paulus ^b, Ming-Qun Xu ^{a,*}

^a New England Biolabs, 32 Tozer Road, Beverly, MA 01915, USA

^b Boston Biomedical Research Institute, Boston, MA 02114, USA

Received 25 October 1996; received in revised form 24 January 1997; accepted 27 January 1997; Received by A. Bernardi

Abstract

A novel protein purification system has been developed which enables purification of free recombinant proteins in a single chromatographic step. The system utilizes a modified protein splicing element (intein) from *Saccharomyces cerevisiae* (*Sce* VMA intein) in conjunction with a chitin-binding domain (CBD) from *Bacillus circulans* as an affinity tag. The concept is based on the observation that the modified *Sce* VMA intein can be induced to undergo a self-cleavage reaction at its N-terminal peptide linkage by 1,4-dithiothreitol (DTT), b-mercaptoethanol (b-ME) or cysteine at low temperatures and over a broad pH range. A target protein is cloned in-frame with the N-terminus of the intein-CBD fusion, and the stable fusion protein is purified by adsorption onto a chitin column. The immobilized fusion protein is then induced to undergo self-cleavage under mild conditions, resulting in the release of the target protein while the intein-CBD fusion remains bound to the column. No exogenous proteolytic cleavage is needed. Furthermore, using this procedure, the purified free target protein can be specifically labeled at its C-terminus. © 1997 Elsevier Science B.V.

Keywords: Protein purification; Intein; Chitin-binding domain; Protein labeling

1. Introduction

Purification of recombinant proteins by the use of affinity tags is a convenient and widely used technology (LaVallie and McCoy, 1995). A target protein is expressed as a fusion with an affinity tag which allows purification on an affinity column. A variety of affinity tags have been used, including *Schistosoma* glutathione

S-transferase (GST) (Smith and Johnson, 1988), *Escherichia coli* maltose-binding protein (MBP) (Guan et al., 1988), *Staphylococcus* protein A (Nilsson and Abrahmsen, 1990), polyhistidine (Van Dyke et al., 1992), and calmodulin-binding peptide (Stofko-Hahn et al., 1992). Following protein purification, the affinity tag is typically cleaved from the target protein by treatment with a site-specific protease (LaVallie and McCoy, 1995). The use of proteases has limited the application of many affinity purification systems. First, cleavage by proteases is not always specific and may result in cleavage at secondary sites within the target protein. Second, the elevated temperatures required for many proteolytic cleavage reactions may unfavorably affect protein stability or activity. Third, the cleavage is sometimes inefficient due to the inaccessibility of the cleavage site on the fusion protein. Finally, additional chromatographic steps are required to separate the target protein from the affinity tag and the protease.

To avoid the limitations of using proteolytic cleavage, we have investigated the utilization of the inducible

* Corresponding author. Tel.: +1 508 9275054, ext. 372; fax: +1 508 9211350; e-mail: xum@neb.com

¹ Present address: Laboratory of Genetics, University of Wisconsin, Madison, WI 53706, USA.

Abbreviations: Amp^r, ampicillin resistance gene; b-ME, b-mercaptoethanol; bp, base pair(s); CBD, chitin-binding domain; DTT, 1,4-dithiothreitol; Intein, modified *Sce* VMA intein; IPTG, isopropyl-b-D-thiogalactopyranoside; kb, kilobase(s) or 1000 bp; kDa, kilodalton(s); *lacI*^q, gene overexpressing *lac* repressor; MBP or M, *E. coli* maltose-binding protein; MCS, multiple cloning site; NEB, New England Biolabs, Inc.; PAGE, polyacrylamide gel electrophoresis; *P*_{tac}, *tac* promoter; PCR, polymerase chain reaction; SDS, sodium dodecyl sulfate.

self-cleavage activity of a protein splicing element in protein purification (Fig. 1A). Protein splicing is a post-translational processing event in which an internal protein segment, the intein (Perler et al., 1994), can catalyze its own excision from a precursor protein and concomitantly ligate the flanking regions, the exteins, to form a mature protein (Kane et al., 1990; Hirata et al., 1990; Davis et al., 1992; Perler et al., 1992; Cooper et al., 1993; Xu et al., 1993). Using in vitro protein splicing systems, we have elucidated the protein splicing pathway of inteins from the thermostable DNA polymerase of *Pyrococcus* sp. GB-D and the 69-kDa vacuolar ATPase subunit of *Saccharomyces cerevisiae* (Xu et al., 1993, 1994; Shao et al., 1995, 1996; Chong et al., 1996; Xu and Perler, 1996). Through amino acid substitution, we have been able to modulate each step in the splicing pathway and convert the splicing reaction into an efficient polypeptide cleavage reaction at either splice junction (Xu and Perler, 1996; Chong et al., 1996). Specifically, a single amino acid substitution, Asn⁴⁵⁴ to Ala, at the C-terminal splice junction of the intein from the 69-kDa vacuolar ATPase subunit of *S. cerevisiae* (*Sce* VMA intein), has been shown to block splicing and C-terminal cleavage, but not the N-terminal cleavage mediated by the N-S acyl shift at Cys1 (Chong et al., 1996) (Fig. 1B). Nucleophiles that react with thioesters, such as thiols (i.e., b-ME, DTT, cysteine) and hydroxylamine, can effectively shift the N-S equilibrium by attacking the thioester, thereby inducing the N-terminal cleavage of the intein. In the case of cysteine-induced cleavage, a spontaneous S-N shift occurs after the nucleophilic attack by the sulfhydryl group of cysteine, resulting in formation of a native peptide bond between the cysteine and the C-terminus of the N-extein (Chong et al., 1996) (Fig. 1B).

Here we report the utilization of the inducible self-cleavage activity of the modified *Sce* VMA intein (Y) to develop a novel protein purification system. We demonstrate the effectiveness of the system by expressing a number of procaryotic and eucaryotic proteins in *E. coli* and purifying them free of their affinity tag after a single chitin column.

2. Results

2.1. Construction of the MYB fusion and characteristics of the CBD as an affinity tag

The maltose-binding protein from *E. coli* (MBP, or M) was fused to the N-terminus of the modified *Sce* VMA intein (Y) containing an Asn⁴⁵⁴Ala substitution, and a small (5 kDa) chitin-binding domain (CBD or B) from the C-terminal region of chitinase A1 from *Bacillus circulans* WL-12 (Watanabe et al., 1994) was fused to the C-terminus of the intein. The resulting fusion

construct, pMYB129, was used as a model system for studies of affinity binding and inducible cleavage at the N-terminal splice junction. The fusion protein MYB bound tightly to chitin beads. In a typical experiment, about 2 mg of the MYB fusion protein from crude cell extract bound to 1 ml of chitin beads (David Landry, NEB, unpublished results). High salt concentrations (up to 1 M NaCl), 0.1% Triton X-100 or Tween 20 and 1 mM EDTA had no effect on the binding of the CBD to chitin beads (data not shown). Buffers containing SDS (>0.5%) or guanidinium (6 M) were required to strip MYB or other CBD fusion proteins from chitin beads (data not shown).

2.2. Thiol-induced cleavage at the intein N-terminus in the MYB fusion protein

To study cleavage of the MYB fusion protein in solution, the unspliced MYB fusion protein (97 kDa) was purified on a single amylose column, with a yield of about 20 mg per liter of culture. The MYB fusion protein was very stable in vivo, during purification and during storage in solution at 4°C. Incubation of MYB at temperatures ranging from 25°C to 50°C resulted in no increase in M and YB (data not shown). The purified MYB fusion protein was treated at 4°C with different concentrations of DTT, b-ME and free cysteine (Fig. 2). More than 90% of MYB was cleaved by 30 mM DTT after 16 h incubation, generating M (42 kDa) and YB (55 kDa) (Fig. 2B), whereas b-ME was less effective (70% cleavage after 16 h; Fig. 2A). Cleavage by DTT was more efficient at high pH (Fig. 2D). Cleavage of MYB by cysteine was less efficient than cleavage by DTT or b-ME and required higher concentrations and higher pH values (Fig. 2C). The cleavage of MYB by hydroxylamine (100 mM) was also tested and found to be less efficient at 4°C than the thiol compounds (data not shown). In the absence of DTT, b-ME or cysteine, the MYB fusion protein was completely stable at 4°C in the 5.5–9.0 pH range. A typical control sample is shown in the first lane of each panel in Fig. 2. Furthermore, the cleavage reaction by DTT, b-ME or cysteine was efficient over a wide range of salt concentrations (50 mM–2 M NaCl) and in the presence of 0.1% Triton X-100 or Tween 20 (data not shown).

2.3. Cleavage of the MYB fusion protein immobilized on chitin beads

The crude cell extract containing MYB was mixed with chitin beads. After extensive wash to remove unbound proteins, different concentrations of b-ME or DTT were applied to the beads to induce the cleavage reaction at 4°C. Only MBP was eluted after a 6 h or 16 h incubation (Fig. 3A and B). These results indicate that cleavage occurred at the intein N-terminus while

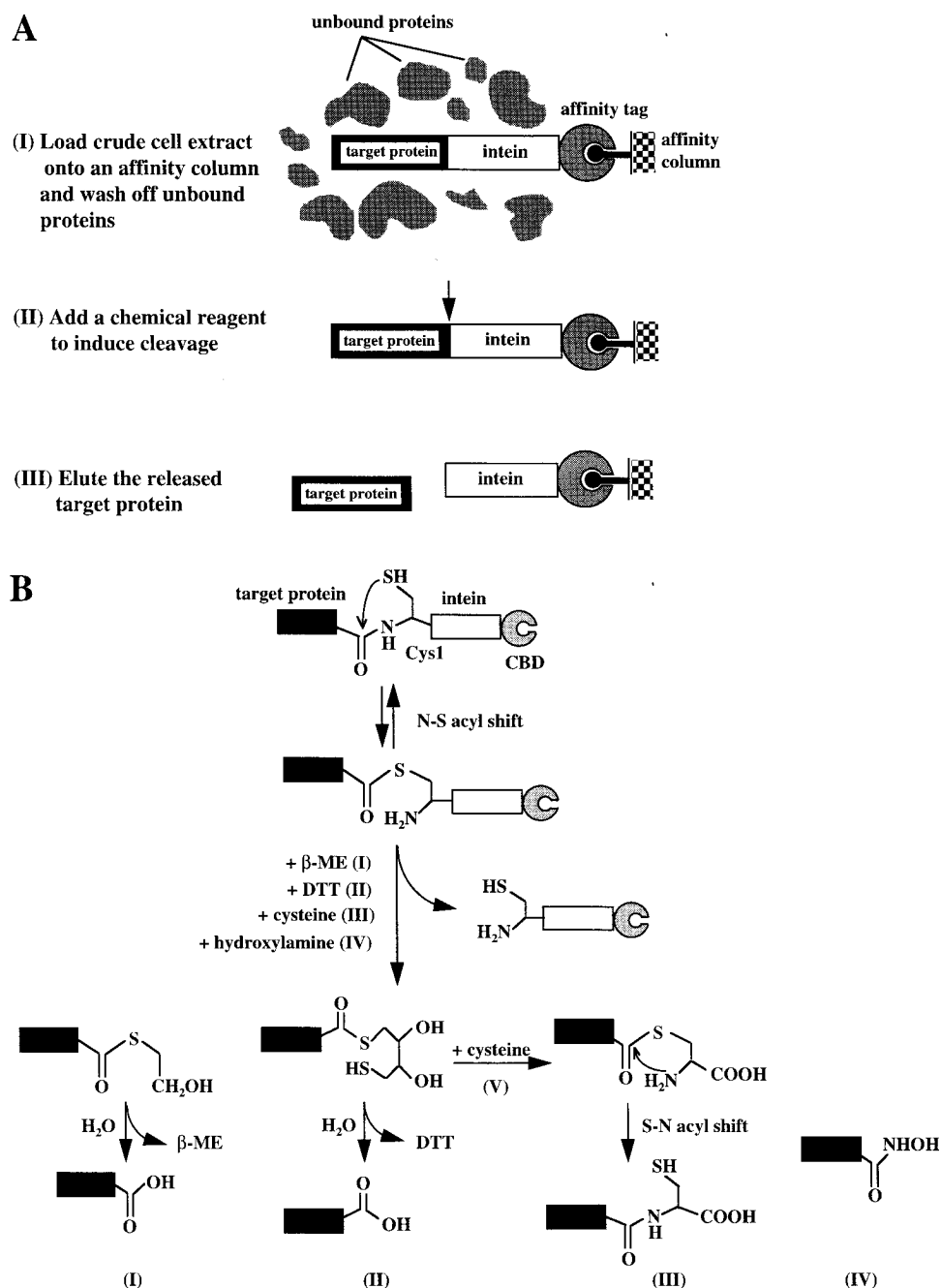


Fig. 1. (A) The concept of the intein-mediated protein purification with a self-cleavable affinity tag. A target protein is fused to the N-terminus of a modified protein splicing element (intein) which is in turn fused to an affinity tag (CBD). The fusion protein from a crude cell extract is purified by adsorption to a chitin affinity column (I). The intein is then induced to undergo on-column self-cleavage (arrow) by a chemical reagent (II). The target protein is specifically released from the column and eluted as a pure protein (III). (B) The chemical mechanism of the intein N-terminal cleavage reaction. The *Scd* VMA intein was modified by an Asn⁴⁵⁴Ala mutation which inactivates its splicing and C-terminal cleavage activities (Chong et al., 1996). This modified intein, when inserted between a target protein and a chitin-binding domain (CBD), catalyzes an N-S acyl shift at its N-terminal Cys1 which results in a thioester bond between the target protein and Cys1. Normally, the equilibrium of the N-S shift strongly favors peptide bond formation and the fusion precursor is quite stable. The presence of β -ME (I), DTT (II), cysteine (III) or hydroxylamine (IV) shifts the N-S equilibrium by cleaving the thioester bond. After cleavage, β -ME (I) or DTT (II) is hydrolyzed from the C-terminus of the target protein, while cysteine (III) or hydroxylamine (IV) remain covalently attached to the target protein. In the case of cysteine, a subsequent S-N shift results in formation of a native peptide bond. Cysteine can also attack the C-terminal thioester bond of the DTT-cleaved target protein and subsequently attach to the target protein through a peptide bond (V). Therefore, when a labeled cysteine is used, the target protein can be specifically labeled at its C-terminus.

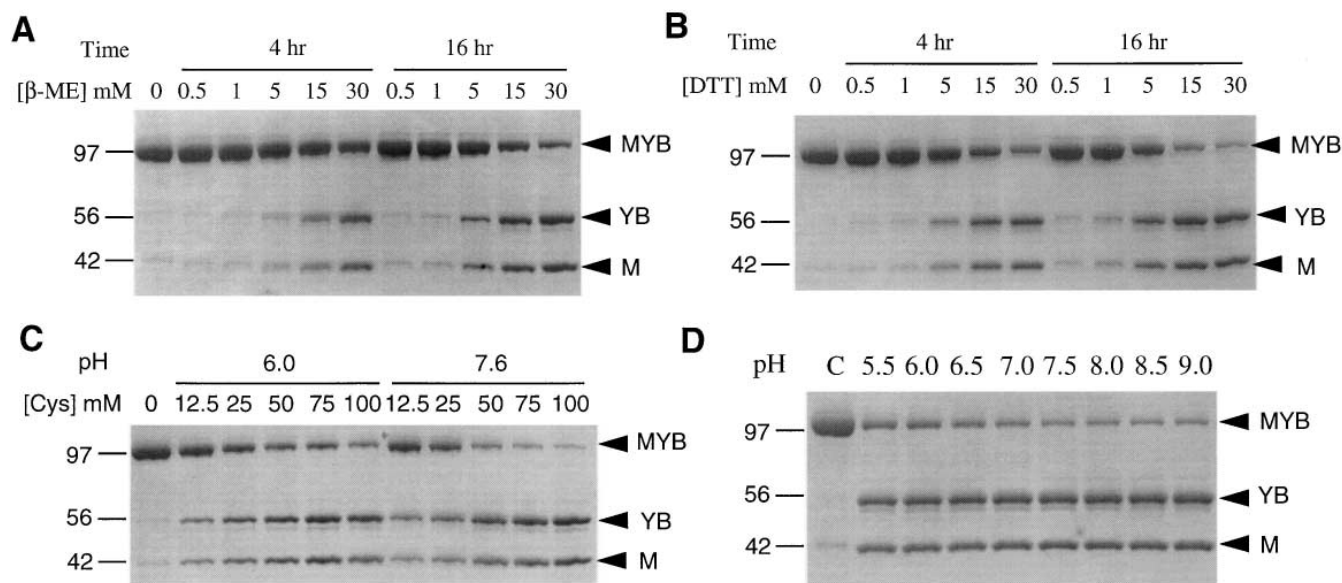


Fig. 2. Induction of cleavage at the intein N-terminus in the MYB fusion protein. Cleavage of purified MYB fusion protein (97 kDa) was induced with b-ME (A), DTT (B) or cysteine (Cys) (C) to generate the maltose-binding protein (M, 42 kDa) and the intein-CBD fusion (YB, 55 kDa). (D) pH effects on MYB cleavage by 30 mM DTT. Protein molecular weight standards (kDa, NEB) are shown on the left side of each panel. A control sample is shown in the first lane of each panel. In panel D, C is a control reaction at pH 8 without DTT. **Methods:** The gene encoding the *B. circulans* CBD was made synthetically using eight oligonucleotides (NEB) comprising both strands of the gene and overlapping by 12 bp. (1) 5'-T ATG ACG ACA AAT CCT GGT GTA TCC GCT TGG CAG GTC-3', (2) 5'-pATA AGC TGT GTT GAC CTG CCA AGC GGA TAC ACC AGG ATT TGT CGT CA-3', (3) 5'-pAAC ACA GCT TAT ACT GCG GGA CAA TTG GTC ACA TAT AAC GGC-3', (4) 5'-pTTT ATA CGT CTT GCC GTT ATA TGT GAC CAA TTG TCC CGC AGT-3', (5) 5'-pAAG ACG TAT AAA TGT TTG CAG CCC CAC ACC TCC TTG GCA GGA-3', (6) 5'-pGGA TGG TTC CCA TCC TGC CAA GGA GGT GTG GGG CTG CAA ACA-3', (7) 5'-pTGG GAA CCA TCC AAC GTT CCT GCC TTG TGG CAG CTT CAA TCG AGC T-3', (8) 5'-CGA TTG AAG CTG CCA CAA GGC AGG AAC GTT-3'. The eight oligonucleotides were annealed and ligated into the *Nde*I and *Sac*I sites of plasmid pMIP22/23, a derivative of pMIP21 (Xu et al., 1993). The resulting plasmid (pBIP22/23) was used as a template for subsequent PCR amplification of the CBD gene. The PCR-synthesized CBD fragment was then used to construct pMIB115, which encodes the gene for a fusion protein consisting of the *E. coli* MBP, the *Psp* pol intein-1 (Xu et al., 1993) and the CBD (F.B. Merisha, unpublished). The modified *See* VMA intein (Y) containing an Asn⁴⁵⁴Ala substitution from pMYT1 (Chong et al., 1996) was inserted in-frame between M and B in pMIB115, replacing the *Psp* pol intein-1. The resulting vector, pMYB129, encodes a gene for the fusion protein MYB. The MYB fusion protein was expressed and purified on amylose resin as described by Chong et al. (1996). The purified MYB fusion protein (1.5 mg/ml) was incubated with various concentrations (0.5–30 mM) of b-ME or DTT in column buffer (20 mM Hepes, pH 8.0; 0.5 M NaCl) for 4 or 16 h at 4°C. For experiments on pH effects, the purified MYB was incubated at 4°C for 16 h with 30 mM DTT in the column buffer at different pH values (5.5–9.0). The N-terminal cleavage of MYB was also induced by treatment of MYB with various concentrations (12.5–100 mM) of cysteine at pH 6.0 or 7.6 in 20 mM Hepes, 0.5 M NaCl for 16 h at 4°C. As controls, the purified MYB was incubated at 4°C at pH 5.5–9.0 for 16 h in the absence of b-ME, DTT or cysteine. All reactions were stopped by mixing the protein samples with 1/2 vol. of 3 × protein sample buffer (NEB) and boiling for 5 min. Each sample (2 ml) was electrophoresed on a 12% Tris-glycine PAGE (Novex). The running buffer was 25 mM Tris, pH 8.3, 190 mM glycine, 0.1% SDS. Staining was with Coomassie Brilliant Blue.

the MYB fusion protein was immobilized on the chitin beads, and that the cleavage conditions had no effect on the binding of the CBD. To examine the efficiency of on-column cleavage, the chitin beads were washed with 2% SDS to strip any remaining MYB fusion protein as well as the cleavage product (YB). More than 90% of the MYB fusion protein was cleaved by 30 mM DTT at 4°C in 16 h (data not shown).

2.4. Construction of the pCYB vectors for cloning of target proteins

The MBP sequence of pMYB129 was replaced with a series of multiple cloning sites, yielding the pCYB vectors (Fig. 4). The pCYB vectors have a ColE1 replica-

tion origin, an M13 origin for generating single-stranded DNA and an ampicillin resistance gene. Protein expression is under control of an IPTG-inducible *tac* promoter. The presence of *lacI*^q repressor gene results in tighter transcriptional control. pCYB vectors with a T7 promoter were also constructed (data not shown). Three multiple cloning sites (MCS) of the pCYB vectors are shown in Fig. 4B. When the *Nde*I site is used for cloning a target gene, translation can initiate at the ATG of the *Nde*I site and the fusion protein will have the natural N-terminus of the target protein. The use of *Sap*I site of MCS1 will result in the fusion of the C-terminus of the target protein to the N-terminal cysteine of the intein (Cys1). Therefore, when the *Nde*I and *Sap*I sites of MCS1 are used, the native sequence of the target

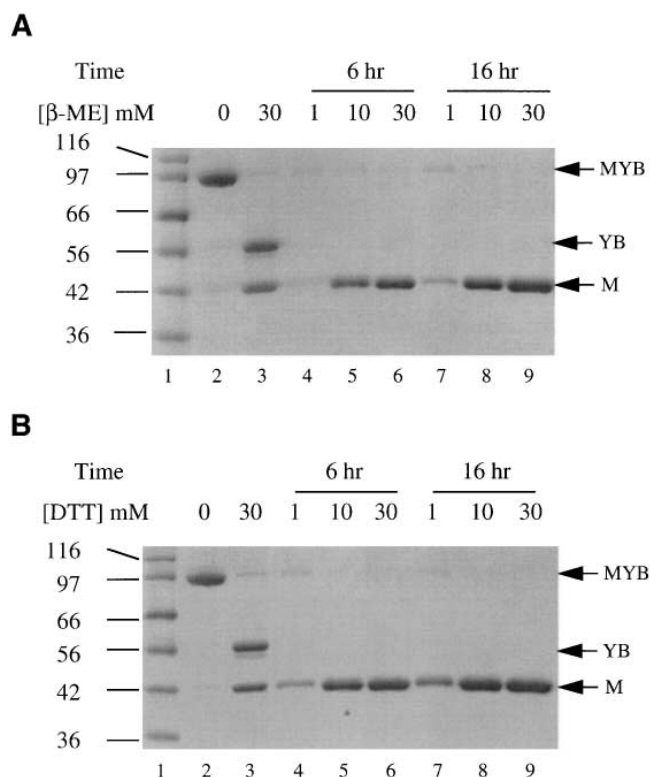


Fig. 3. Induction of cleavage of MYB immobilized on chitin beads with b-ME (A) or DTT (B). Lanes: 1, protein molecular weight standards (kDa, NEB); 2, purified MYB; 3, cleavage of the purified MYB in solution by 30 mM b-ME or DTT at room temperature for 6 h; 4–9, eluate from chitin beads after treatment of bound MYB protein with different concentrations of b-ME or DTT at 4°C for 6 or 16 h. The samples were examined by Coomassie Blue staining. M, maltose-binding protein; YB, intein-CBD fusion. **Methods:** Spherical chitin beads were prepared by emulsification of a solution of chitosan in 10% acetic acid with an organic solvent containing a suitable surfactant, followed by the addition of acetic anhydride (Roberts and Taylor, 1988). After gelation, the resulting beads (50–100 µm) were washed with water. Following dehydration by washing with methanol, the beads were reacylated with acetic anhydride. The degree of *N*-acetylation was determined with the 2,4,6-trinitrobenzenesulfonic acid and perchloric acid assay using a glucosamine standard (Wilkie and Landry, 1988). Chitin beads are commercially available from NEB (cat. No. 6651). The MYB fusion protein was expressed as described by Chong et al. (1996). 1 ml of the clarified cell extract was mixed with 0.2 ml chitin beads in an Eppendorf tube. After 10 min incubation at 4°C, the beads were washed 5 times with column buffer. b-ME or DTT at various concentrations (1–30 mM) in the column buffer was then applied to the beads and the samples were incubated at 4°C for 6 or 16 h. Chitin beads were spun down and the supernatant was collected and analyzed by SDS-PAGE as described in Fig. 2.

protein will be obtained after cleavage, with no vector-derived amino acid residues. When a target gene is cloned into the *Nde*I and *Sma*I sites of MCS2, the target protein will have an additional glycine residue at its C-terminus after cleavage. When the *Nde*I and *Xho*I sites of MCS3 are used, the target protein will have three additional residues (leucine-glutamate-glycine) at its C-terminus.

2.5. Expression and purification of target proteins using pCYB vectors

To demonstrate that the pCYB vectors can be used to express and purify recombinant proteins, a number of proteins from both procaryotes and eucaryotes were cloned into the pCYB vectors and purified after a single chitin column as summarized in Table 1. The procedures for the expression and purification of *Hha*I methylase (Caserta et al., 1987) are illustrated in Fig. 5A. After induction with IPTG, the fusion protein (approx. 95 kDa) could be observed in the crude extract (Fig. 5A, lane 3), but not in uninduced cells (Fig. 5A, lane 2). After the crude extract from the induced cells was loaded onto a chitin column, most of the fusion protein bound to the chitin resin (Fig. 5A, lane 4). The column was then washed with column buffer. DTT (30 mM) was then quickly passed through the column to induce the on-column cleavage reaction. No significant amount of *Hha*I methylase was cleaved off the column during this brief DTT flush (Fig. 5A, lane 5). The column was then incubated at 4°C overnight, followed by elution of released *Hha*I methylase using the column buffer without DTT (Fig. 5A, lanes 6–9). The enzyme was more than 95% pure as estimated by staining an SDS-PAGE gel with Coomassie Blue and was active as determined by the standard methylase assay (Caserta et al., 1987) (data not shown). Washing the column with 2% SDS to strip the bound protein indicated that more than 90% of the fusion protein was cleaved (Fig. 5A, lane 10). The expression and purification of some of the recombinant proteins are shown in Fig. 5B. The specific activities of the purified *Bst* DNA polymerase large fragment (Aliotta et al., 1996) and the *Bam*HI, *Bgl*II, and *Eco*RV restriction endonucleases (Brooks et al., 1989; Kravets et al., 1990) were determined and found to be more than 70% of the values obtained with enzymes purified by conventional methods (Table 1). The yield of the *Bst* DNA polymerase large fragment was increased by 4-fold if a T7 promoter was used in place of a *tac* promoter. The N-terminal catalytic domain (residues 1–177) of *Brugia malayi* cyclophilin PPIase A (Page et al., 1995) was purified and a minor band of approx. 55 kDa corresponding to the intein-CBD fusion partner was also present in the eluate. The purified calmodulin-dependent protein kinase (CamKII) (Lin et al., 1987) showed an calmodulin-dependent protein kinase activity similar to the wild-type CamKII purified by a eucaryotic expression system, albeit with reduced activity (S.T. Wong-Madden, NEB, personal communication). The CamKII sample also contained a minor species migrating slightly slower than the major CamKII band. The purified T4 DNA ligase exhibited 90% of the specific activity of the enzyme purified by multiple chromatographic columns (Charles Card, NEB, personal communication).

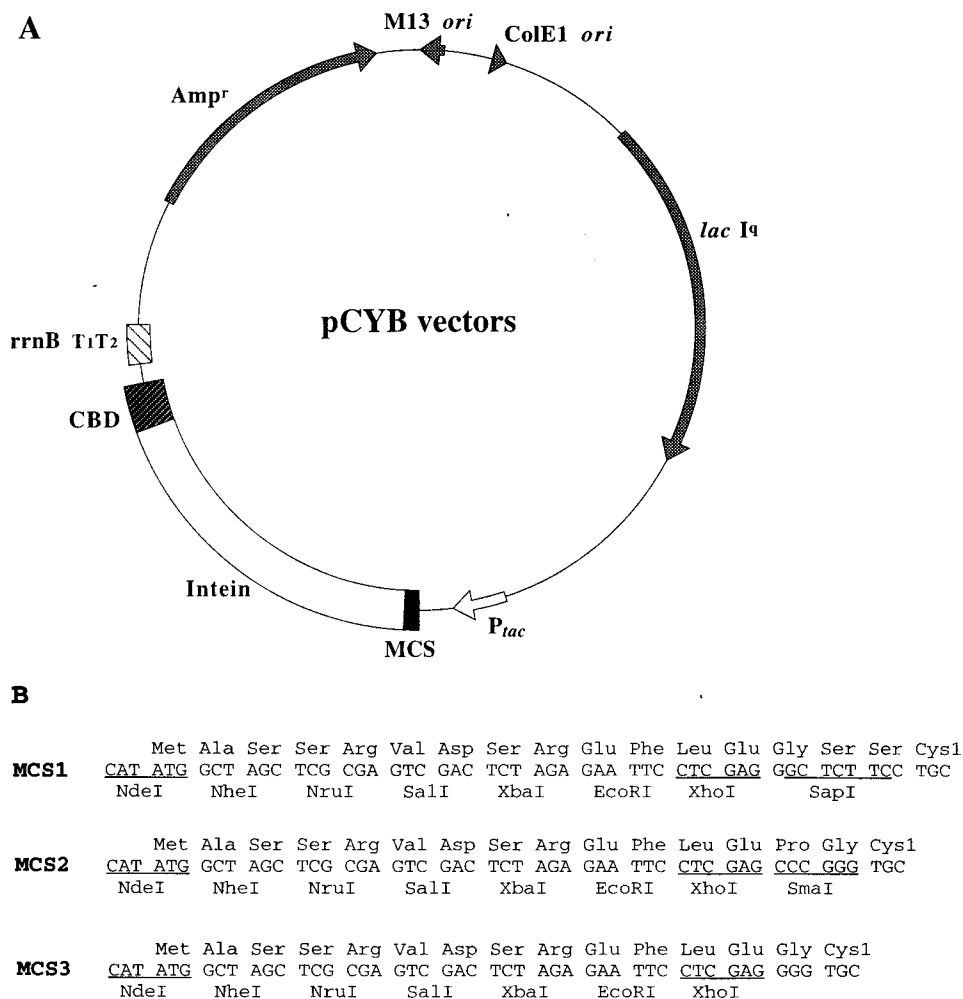


Fig. 4. (A) The pCYB *E. coli* expression vectors. All pCYB vectors contain the modified *See* VMA intein (Y, 1362 bp), carrying an Asn⁴⁵⁴Ala substitution, and the *B. circulans* chitin-binding domain (CBD or B, 135 bp). The multiple cloning sequence (MCS) was designed for insertion of the target gene in-frame with the downstream *See* VMA intein-CBD coding region. Expression of intein-CBD fusion proteins is under control of a *tac* promoter and regulated by the *lacI^q* repressor. The *rrnB* transcription terminators (*rrnBT₁T₂*) are located downstream from CBD. The pCYB vectors have a *ColE1* replication origin, an M13 origin for generating single-stranded DNA and an ampicillin resistance gene. (B) Multiple cloning sites (MCS) of the pCYB vectors. Translation initiates at the ATG of the *NdeI* site. The amino acid sequence is shown above the nucleotide sequence and restriction enzyme sites are indicated below the nucleotide sequence. Cys1, the first (N-terminal) amino acid residue of the intein and the site of cleavage, is not part of the multiple cloning site. The recognition sequences of *NdeI*, *SapI*, *SmaI* and *XhoI* are underlined. The ATG sequence of the *NdeI* site in pMIB124, a derivative of pMIB115, is 9 bp downstream from the ribosome-binding site of the *E. coli* *malE* gene (Guan et al., 1988) and is used as a translation start codon. **Methods:** pMIB124 was constructed by replacing the *E. coli* MBP coding region in pMIB115 with a polylinker containing *NdeI* and *XhoI* sites. pMIB124 was digested by *XhoI* and *PstI* and ligated with the 1.5 kb *XhoI*-*PstI* fragment containing the *See* VMA intein-CBD coding region from pMYB129 replacing the *Psp* pol intein-1-CBD sequence. The resulting plasmid pMYB131 was digested by *NdeI*-*XhoI* and ligated with a linker containing MCS3 to yield pCYB138 (6840 bp). pCYB1 and pCYB2 vectors, with MCS1 and MCS2 respectively, were created by introducing *SapI* and *SmaI* site, respectively, through linker substitution of the region (flanking the N-terminus of the intein) between *XhoI* and *KpnI* sites of pCYB138.

2.6. *In vitro* labeling of MBP and BamHI

Following DTT-induced on-column cleavage, purified proteins were tested for replacement of the C-terminal DTT moiety by radiolabeled cysteine. The initial nucleophilic attack by the sulfhydryl group of cysteine followed by a spontaneous rearrangement (S-N shift) results in a peptide bond between the cysteine and the C-terminus of the target protein (Chong et al., 1996; Dawson et al.,

1994) (Fig. 1B). Aliquots of MBP and *Bam*HI eluted from the chitin beads by DTT-induced cleavage were incubated with L-[³⁵S]cysteine (at 1 mM final concentration) immediately after elution. As shown in Fig. 5B, MBP (MBP panel: lane 5) and *Bam*HI (*Bam*H panel: lane 4) were both labeled. On the other hand, when a sample of MBP, purified on amylose resin (NEB) by elution with maltose, was incubated with L-[³⁵S]cysteine, no labeling was observed (MBP panel: lane 4).

Table 1
Recombinant proteins expressed and purified by pCYB vectors

Target protein ^a	Molecular weight (kDa)	Cloning sites	Promoter	Host <i>E. coli</i> strain ^b	Final yield (mg/l culture)	Activity ^a
<i>HhaI</i> methylase	40	<i>NdeI</i> , <i>XhoI</i> (MCS3)	<i>tac</i>	ER2267	3	+
MBP	42	<i>NdeI</i> , <i>XhoI</i> (MCS3)	<i>tac</i>	ER2267	20	n.d.
<i>Bst</i> DNA polymerase large fragment	65	<i>NdeI</i> , <i>XhoI</i> (MCS3)	T7	ER2504	2	90%
			<i>tac</i>	ER2504	0.5	+
T4 DNA ligase	60	<i>NdeI</i> , <i>SmaI</i> (MCS2)	<i>tac</i>	ER2267	4	90%
<i>BamHI</i> ^d	26	<i>NdeI</i> , <i>SapI</i> (MCS1)	T7	ER2504	1	75%
<i>BglII</i> ^d	24	<i>XbaI</i> , <i>XhoI</i>	T7	ER2504	4	99%
<i>EcoRV</i> ^d	26	<i>NdeI</i> , <i>XhoI</i> (MCS3)	T7	ER2504	0.5–1	72%
Cyclophilin PPIase	23	<i>NdeI</i> , <i>XhoI</i> (MCS3)	<i>tac</i>	ER2267	0.5	+
CamKII	33	<i>NdeI</i> , <i>SapI</i> (MCS1)	T7	ER2504	0.8	+
			<i>tac</i>	ER2504	0.8	+

^aAbbreviations are as in the legend to Fig. 5. The methods for cloning are described in Fig. 5.

^bER2267 (Fomenkov et al., 1994) is F[−] e14[−] (McrA[−]) *end* A1 SupE44 *thi*-1 D(malB) D(*argF-lac*)U169 Pro⁺zjc::Tn5(Kan^r) *fhuA2*. ER2504 is a derivative of BL21(DE3) (Studier et al., 1990) (W. Jack and E.A. Raleigh, NEB, unpublished).

^cThe specific activity of each enzyme determined by an appropriate assay is shown as the percentage of the specific activity of the conventionally purified enzyme. In cases where the specific activities were not measured quantitatively, a plus (+) indicates that the enzyme showed activity after purification. n.d., the activity was not determined.

^dThe toxicity of even low levels of restriction endonuclease required the use of pCYB vectors with a T7 promoter and the co-expression of a corresponding methylase gene.

3. Discussion

In this paper, we describe a novel protein purification system which utilizes the controlled N-terminal cleavage activity of a modified protein splicing element. Unlike other protein purification methods employing affinity tags, this system allows a single-column purification of free recombinant proteins from crude cell lysates through a unique on-column cleavage reaction induced by simple and inexpensive chemical reagents. The intein-mediated cleavage separates the target protein from the affinity tag and eliminates the need for exogenous proteolytic cleavage and chromatographic removal of the protease and affinity tag from the target protein.

The key component of this system is the *Sce* VMA intein modified to undergo cleavage reaction only at its N-terminus. The MYB fusion protein was employed as a model system to study the cleavage reaction. A unique feature of this modified *Sce* VMA intein is that the cleavage reaction can be induced at 4°C (a preferred temperature for purification of proteins) (Fig. 2). Basic pH values favored DTT-induced cleavage, probably owing to the fact that the cleavage reaction is a nucleophilic substitution involving thiolate anion as the nucleophile. However, after 16 h incubation at 4°C, cleavage was mostly complete at all pH values ranging from 5.5 to 9.0 (Fig. 2D). This can be an advantage for purifying proteins with different pH-stability profiles. Furthermore, the ability to carry out cleavage reactions at high salt concentrations or in the presence of non-ionic detergents may improve the solubility of some target proteins. b-ME and DTT at concentrations up to 5 mM did not result in significant cleavage of MYB

protein (Fig. 2), and target proteins that are sensitive to oxidation can therefore be purified in the presence of low concentrations of thiol reagents. After inducing cleavage, both b-ME and DTT formed a thioester bond with the C-terminus of the target protein (Fig. 1B). In the case of the MYB fusion protein, more than 90% of the b-ME or DTT moiety was hydrolyzed from the eluted MBP after 1 week storage at 4°C, pH 8.0, resulting in an unmodified MBP with a native C-terminus (data not shown). The intein-mediated cleavage reaction is highly specific and the target protein should therefore have a homogeneous C-terminus. This was supported by mass spectrometric analysis and sequencing of the C-terminus of the MBP domain after thiol-induced cleavage (Chong et al., 1996). Cysteine or hydroxylamine can also induce N-terminal cleavage reaction, resulting in a target protein with a modified C-terminus (Fig. 1B). For target proteins sensitive to reducing reagents (e.g., owing to the presence of intramolecular disulfide bonds), hydroxylamine can be used for inducing cleavage instead of b-ME and DTT.

To facilitate the purification of fusion proteins, the chitin-binding domain (CBD) from *B. circulans* was incorporated into the system. The CBD has an extremely high affinity for the chitin beads, which allows better recovery of the fusion protein from the crude cell extract. In addition, stringent wash conditions (e.g., high salt concentration or use of detergents) can be employed to reduce non-specific binding, thus increasing purity.

Another unique feature of this system is its ability to allow specific labeling of the C-terminus of a target protein. Since the labeled cysteine only reacts with the C-terminal DTT moiety of the purified target protein,

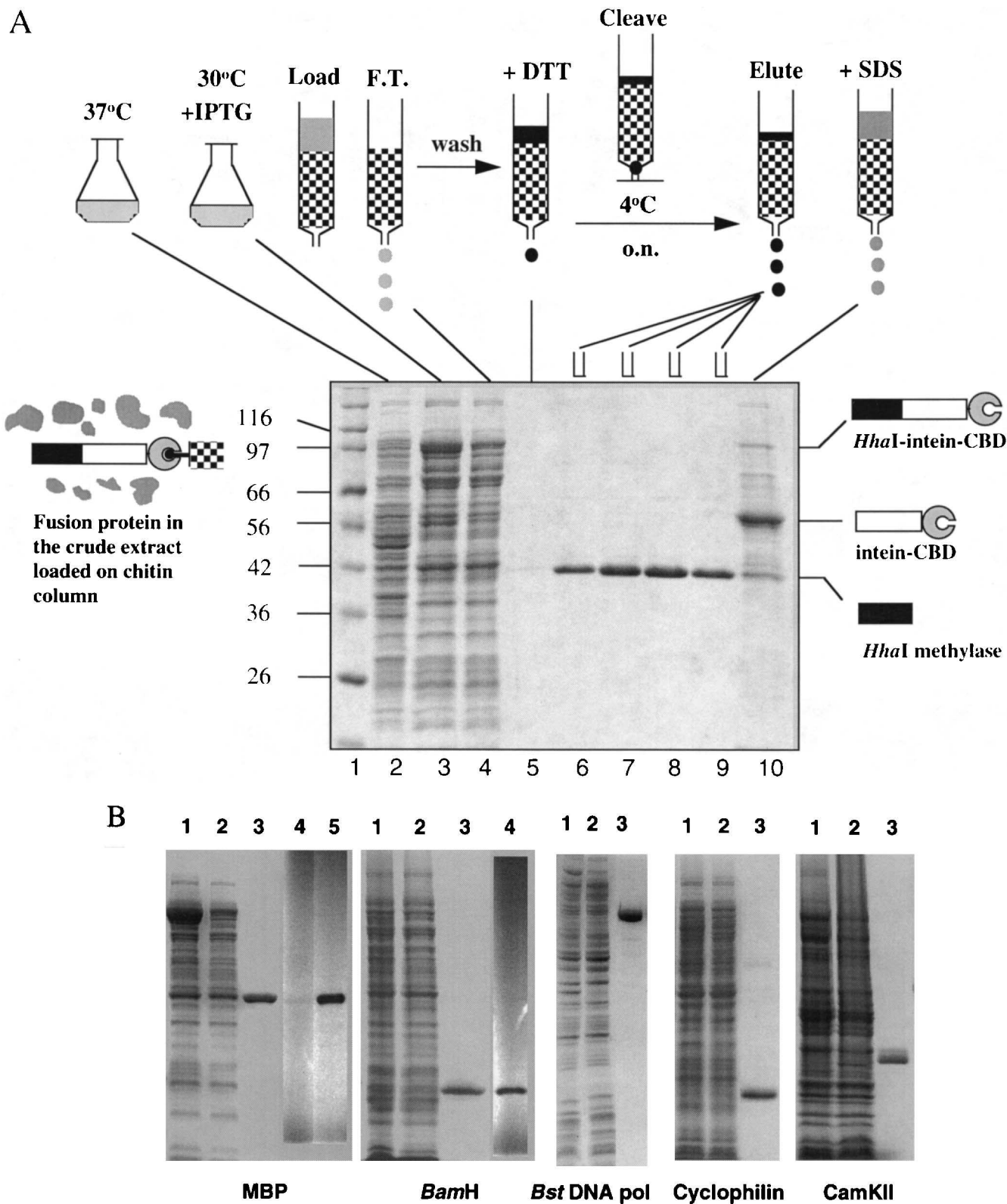


Fig. 5. Single column purification of recombinant proteins from *E. coli* using pCYB vectors. (A) Schematic representation of the expression and purification of *HhaI* methylase using a pCYB vector. Samples taken from different steps during the expression and purification procedures were separated by SDS-PAGE and the gel was stained with Coomassie Blue. Lanes: 1, protein molecular weight standards (kDa, NEB); 2, uninduced cell extract; 3, induced cell extract; 4, flow through (F.T.) from the load. After loading, the column is washed with column buffer until the protein content of the eluate reached a minimum; 5, flow through from the quick DTT flush; 6–9, the first four fractions of the elution after the 4°C overnight (o.n.) incubation in the presence of DTT; 10, a fraction from the SDS elution. (B) Purified proteins after a single chitin column. MBP, *E. coli* maltose-binding protein (42 kDa); *BamHI*, a *Bacillus amyloliquefaciens* H restriction endonuclease (26 kDa); *Bst* DNA pol, *Bacillus stearother*

this method of protein labeling is more specific, less laborious and less likely to affect the structure and activity of the target protein than other methods of in vitro protein labeling (Shumacher and Tsomides, 1996). The utilization of purified proteins labeled in vitro at their C-termini may facilitate the investigation of their biological functions by newly developing techniques such as protein footprinting (Shumacher and Tsomides, 1996). In addition, this system provides an attractive alternative method for peptide synthesis and purification. Furthermore, the thiol esters that result from intein-mediated cleavage induced by thiol compounds can serve as intermediates in peptide ligation (Dawson et al., 1994).

Like other commonly used expression vectors, the pCYB vectors use a *tac* or T7 promoter for expression of heterologous proteins in *E. coli*. Other protein fusion

systems can sometimes increase the expression of the fusion protein by placing a highly expressed affinity tag at the N-terminus of the fusion (LaVallie and McCoy, 1995; Smith and Johnson, 1988; Guan et al., 1988) (e.g., MBP, GST, etc.). In the pCYB vectors, however, translation of the fusion protein starts with the target protein, and therefore the expression level of the fusion protein is primarily determined by the expression of the target protein itself. We have found that the level of protein expression in the pCYB vectors, like T7 promoter vectors, varies with different target proteins. Using a T7 promoter often improves the expression of poorly expressed proteins (Studier et al., 1990). However, some target proteins, especially those from eucaryotes, are expressed poorly in the pCYB vectors (data not shown). In the case of CamKII, the purified protein also had a lower activity compared to the enzyme purified by a

mophilus DNA polymerase large fragment (68 kDa); Cyclophilin, *Brugia malayi* cyclophilin PPIase N-terminal domain (23 kDa); CamKII, a calmodulin-dependent protein kinase (33 kDa). For each panel, lanes: 1, crude cell extract; 2, flow through; 3, a fraction after DTT-induced cleavage; for MBP, 4, control sample for in vitro labeling using amylose-purified MBP; 5, in vitro labeling of chitin-purified MBP with L-[³⁵S]cysteine; for *Bam*HI, lane 4, in vitro labeling with L-[³⁵S]cysteine. **Methods:** All target genes were amplified by PCR to include *Nde*I site at their N termini. The gene for *Hha*I methylase was amplified by PCR from pNW2801 (Caserta et al., 1987) (a gift of Dr. Geoffrey Wilson, NEB). The forward primer, 5'-GGC GGC CAT ATG ATT GAA ATA AAA GAT AAA CAG CTC-3', contains an *Nde*I site (underlined) where the translation initiates at the ATG codon, and the first eight codons of the *Hha*I methylase gene (bold). The reverse primer, 5'-GGC GGC CTC GAG ATA TGG TTT GAA ATT TAA TGA TGA ACC-3', contains an *Xho*I site (underlined) and the sequence of the last 9 amino acids of the *Hha*I methylase gene (bold). PCR mixtures (100 µl) contained Vent DNA polymerase buffer (NEB), 4 mM magnesium sulfate, 400 mM of each dNTP, 1 mM of each primer, 50 ng pNW2801 DNA and 1.0 unit of Vent DNA polymerase (NEB). Amplification was carried out using a Perkin-Elmer/Cetus thermal cycler at 94°C for 1 min, 50°C for 1 min and 72°C for 2 min, for 22 cycles. The PCR products were gel-purified and, after double digestion with *Nde*I and *Xho*I, directly ligated to a gel-purified *Nde*I-*Xho*I double-digested pCYB vector carrying a *tac* promoter and MCS3 to yield pCYB(MHha). *E. coli* strain ER2267 was transformed by pCYB(MHha) and the cells were grown and induced as described below for pMYB129. Similar procedures were followed in the cloning and expression of proteins shown in B and Table 1. For MBP, the expression vector was pMYB129. A 1-l culture was induced at 30°C by 0.3 mM IPTG for 4 h. For *Bst* DNA polymerase large fragment, the cell culture (1 l) was induced at 30°C for 4 h. For *Bam*HI, a 100-l culture was grown in a New Brunswick Magnaferm fermentor at 37°C and then induced at 30°C for 4 h. The pCYB vectors with a T7 promoter contain an *Xba*I site between the promoter and the ribosome-binding site. A PCR-generated fragment containing the gene for *Bgl*II (Anton et al., 1997), the ribosome-binding site and translation start codon (ATG) was cloned into the *Xba*I and *Xho*I sites of a pCYB vector with a T7 promoter, and the cells (1 l) were induced at 15°C for 16 h. For *Eco*RV, the culture (1 l) was induced at 15°C for 16 h. Expression of restriction endonucleases *Bam*HI (Brooks et al., 1989), *Bgl*II or *Eco*RV (Kravets et al., 1990) requires the use of pCYB vectors with a T7 promoter and the co-expression of a corresponding methylase gene in order to prevent the endonuclease from being too toxic to the cells. In the case of *Bam*HI or *Bgl*II, a compatible plasmid (a pACYC184 derivative, NEB) containing the corresponding methylase gene was co-transformed into the cells. In the case of *Eco*RV, the methylase gene was cloned into the same vector as the endonuclease gene but under its own promoter. The coding sequence of the N-terminal catalytic domain (residues 1–177) of *Br. malayi* cyclophilin PPIase was amplified from pMAL-CYP10 (a gift of Clotilde Carlow, NEB) and cloned into pCYB138, yielding pCYP127. The cells harboring pCYP127 (1 l) were induced at 15°C for 16 h. The gene for a calmodulin-dependent protein kinase (CamKII) was amplified by PCR from pSRalpha-CaMK (a gift of Dr. Howard Shulman), and the cells (1 l) were induced at 15°C for 16 h. The coding sequence of T4 DNA ligase (Armstrong et al., 1983) was amplified from a wild-type phage T4D strain (a gift of Dr. David Shub) and cloned into MCS2 (Table 1), yielding pT4YB1. ER2267 cells bearing pT4YB1 were induced by 0.3 mM IPTG at 15°C for 16 h. All purification procedures were conducted at 0–4°C. The cell pellet (3–5 g) from each 1-l culture was resuspended in 50 ml of column buffer (20 mM Hepes, pH 8.0; 0.5 M NaCl) and broken by sonication. After centrifugation at 25 000 × g for 30 min, the supernatant was passed through a column packed with 20 ml of chitin beads (NEB) at a flow rate of 0.5 ml/min and the column was then washed with the column buffer at a flow rate of 2 ml/min until the protein content of the eluate reached a minimum (OD_{595nm} < 0.05 as measured by Bradford assay). Column buffer (50 ml) containing 30 mM DTT was quickly passed through the column in order to distribute DTT evenly throughout the resin and the flow was stopped. The column was incubated at 4°C overnight. Fractions (3 ml each) containing the target protein were obtained by eluting the column with the column buffer in the absence of DTT. After elution, 50 ml of 2% SDS in the column buffer were passed through the column at room temperature to strip the remaining bound proteins and fractions (4 ml each) were collected. The samples were analyzed by SDS-PAGE as described in Fig. 2. The results for *Hha*I methylase are shown in A. The expression and purification of *Bam*HI was similar but conducted on a production scale. The cell pellet (120 g) was resuspended in 500 ml column buffer before sonication and the crude cell extract was passed through a 180 ml column of chitin beads (NEB). The DTT-induced cleavage was conducted at 4°C for 1–3 days. For in vitro labeling of proteins, MBP (1 mg/ml) or *Bam*HI (0.5 mg/ml) was obtained by induction of on-column cleavage of the fusion proteins as described above. An aliquot (36 ml) of each freshly eluted sample was mixed with 4 ml of L-[³⁵S]cysteine (11.0 mCi/ml, 0.0102 mmol/ml, Du Pont NEN) and incubated at 4°C overnight. In a control reaction, an aliquot (36 ml) of MBP (1 mg/ml) purified on an amylose resin column (NEB) was incubated with 4 ml of L-[³⁵S]cysteine. Samples were electrophoresed on a 12% Tris-glycine PAGE (Novex) visualized by autoradiography.

eucaryotic system. This could be due to protein misfolding or lack of post-translational modifications in *E. coli*, or sensitivity to the high concentration of thiol reagent during purification. The development of an analogous eucaryotic expression system may help to improve the expression and activities of eucaryotic target proteins.

The level of protein expression in the pCYB vectors was also affected by induction temperature. We found that in most cases (e.g., *Bst* pol, *Bam*HI, *Bgl*II, *Eco*RV), fusion protein expression was higher at lower induction temperatures (15–30°C) than at 37°C (data not shown). However, in some cases (e.g., MBP), the induction temperature made no difference in expression. Thus, different induction conditions should be tested for each target protein in order to achieve optimal induction. Due to the specificity of the cleavage reaction and high affinity of CBD for chitin beads, pCYB vectors usually resulted in highly purified target proteins. However, when the expression of the target protein was inefficient, significant amounts of other proteins were co-purified (Fig. 5B). In the case of cyclophilin and CamKII, whose expression and final yield were relatively low (Table 1), some impurities could be seen on SDS-PAGE, probably as a result of protease degradation, unspecific binding to chitin beads, or elution of the fusion protein or fusion partner (i.e., intein-CBD fusion).

pCYB vectors are N-terminal fusion vectors in which the C-terminus of a target protein is fused to the N-terminus of the intein. The structure or sequence at the C-terminus of the target protein may affect cleavage efficiency. Preliminary data indicate that some residues, when placed immediately adjacent to Cys1, can either block the cleavage reaction or cause in vivo cleavage (data not shown). Therefore, in some cases, the addition of a favorable residue or sequence between the target protein and the intein may be necessary for the controllable cleavage reaction (e.g., using pCYB vectors with MCS2 and MCS3 for cloning). Furthermore, incomplete processing of N-terminal methionine in some fusion proteins may generate a target protein with heterogeneous N-termini. Nevertheless, the method described here clearly has many advantages and has been shown to drastically simplify protein purification. Using the pCYB vectors, many proteins were successfully purified with just one column with a yield comparable to or greater than the yield using conventional methods. Furthermore, most of the tested target proteins retain their activities after purification (Table 1).

In conclusion, by combining a modified protein splicing element with an affinity tag in protein purification, we have been able to obtain highly purified non-tagged recombinant proteins in a single chromatographic step. Owing to the self-catalytic property of the intein, this purification system does not require any cellular co-factor for cleavage, and the core components of this system (i.e., intein-CBD) should therefore be compatible

with other expression and purification systems (e.g., yeast, baculovirus, mammalian cells, etc.).

Acknowledgement

We thank Ira Schildkraut, Richard Roberts, Yang Shao, William Jack, Christopher Noren, Paul Riggs, Elisabeth Raleigh, Maurice Southworth, Yi Tan, Michael Comb, Daniel Panne and Jill Platko for valuable discussions and reading of the manuscript; Geoffrey Wilson for the gift of pNW2801 containing *Hha*I methylase gene; William Jack, Richard Morgan, Jurate Bitinaite for the clones of restriction endonucleases and methylases; Gloria-Jane Healy, Richard Whitaker, Sharon Wong-Madden, Diana T. O'Loane, Dong Ma, Tao Cui, Robyn Byer and Charles Card for technical assistance; David Shub for providing phage T4D strain; Howard Shulman for plasmid pSRalpha-CaMK; Huimin Kong for the *Bst* DNA polymerase large fragment clone; and Clotilde Carlow for the clone of the *Br. malayi* cyclophilin PPIase.

References

- Aliotta, J.M., Pelletier, J.J., Ware, J.L., Moran, L.S., Benner, J.S., Kong, H., 1996. Thermostable *Bst* DNA polymerase I lacks a 3'→5' proofreading exonuclease activity. *Genet. Anal. Biomol. Eng.* 12, 185–195.
- Anton, B.P., Heiter, D.F., Benner, J.S., Hess, E.J., Greenough, L., Moran, L.S., Slatko, B.E., Brooks, J.E., 1997. Cloning and characterization of the *Bgl*II restriction-modification system reveals a possible evolutionary footprint. *Gene* 187, 19–27.
- Armstrong, A., Brown, R.S., Tsugita, A., 1983. Primary structure and genetic organization of phage T4 DNA ligase. *Nucleic Acids Res.* 11, 7145–7156.
- Brooks, J.E., Benner, J.S., Heiter, D.F., Silber, K.R., Sznyter, L.A., Jager-Quinton, T., Moran, L.S., Slatko, B.E., Wilson, G.G., Nwankwo, D.O., 1989. Cloning the *Bam*HI restriction-modification system. *Nucleic Acids Res.* 13, 979–997.
- Caserta, M., Zacharias, W., Nwankwo, D., Wilson, G.G., Wells, R.D., 1987. Cloning, sequencing, in vivo promoter mapping and expression in *Escherichia coli* of the gene for the *Hha*I methyltransferase. *J. Biol. Chem.* 262, 4770–4777.
- Chong, S., Yang, S., Paulus, H., Benner, J., Perler, F.B., Xu, M.-Q., 1996. Protein splicing involving the *Saccharomyces cerevisiae* VMA intein: the steps in the splicing pathway, side reactions leading to protein cleavage, and establishment of an in vitro splicing system. 1996. *J. Biol. Chem.* 271, 22159–22168.
- Cooper, A.A., Chen, Y., Lindorfer, M.A., Stevens, T.H., 1993. Protein splicing of the yeast TFP1 intervening protein sequence: a model for self-excision. *EMBO J.* 12, 2575–2583.
- Davis, E.O., Jenner, P.J., Brooks, P.C., Colston, M.J., Sedgwick, S.G., 1992. Protein splicing in the maturation of *M. tuberculosis* RecA protein: a mechanism for tolerating a novel class of intervening sequence. *Cell* 71, 201–210.
- Dawson, P.E., Muir, T.W., Clark-Lewis, I., Kent, S.B.H., 1994. Synthesis of proteins by native chemical ligation. *Science* 266, 776–779.
- Fomenkov, A., Xiao, J.-P., Dila, D., Raleigh, E., Xu, S.-Y., 1994. The

- 'endo-blue method' for direct cloning of restriction endonuclease genes in *E. coli*. Nucleic Acids Res. 22, 2399–2403.
- Guan, C. di, Li, P., Riggs, P.D., Inouye, H., 1988. Vectors that facilitate the expression and purification of foreign peptides in *Escherichia coli* by fusion to maltose-binding protein. Gene 67, 21–30.
- Hirata, R., Ohsumi, Y., Nakano, A., Kawasaki, H., Suzuki, K., Anraku, Y., 1990. Molecular structure of a gene, VMA1, encoding the catalytic subunit of H⁺-translocating adenosine triphosphatase from vacuolar membranes of *Saccharomyces cerevisiae*. J. Biol. Chem. 265, 6726–6733.
- Kane, P.M., Yamashiro, C.T., Wolczyk, D.F., Neff, N., Goebel, M., Stevens, T.H., 1990. Protein splicing converts the yeast TFP1 gene product to the 69-kD subunit of the vacuolar H⁺-adenosine triphosphatase. Science 250, 651–657.
- Kravets, A.N., Zakharova, M.V., Solonin, A.S., Kuz'min, N.P., Taniashin, V.I., Glatman, L.I., Moroz, A.F., Baev, A.A., 1990. Cloning and regulation of gene expression of *EcoRV* restriction-modification system. Mol. Biol. 24, 438–447.
- LaVallie, E.R., McCoy, J.M., 1995. Gene fusion expression systems in *Escherichia coli*. Curr. Opin. Biotechnol. 6, 501–506.
- Lin, C.R., Kapiloff, M.S., Durgerian, S., Tatemoto, K., Russo, A.F., Hanson, P., Schulman, H., Rosenfield, M.G., 1987. Molecular cloning of a brain-specific calcium/calmodulin-dependent protein kinase. Proc. Natl. Acad. Sci. USA 84, 5962–5966.
- Nilsson, B., Abrahmsen, L., 1990. Fusions to Staphylococcal protein A. Methods Enzymol. 185, 144–161.
- Page, A.P., Landry, D., Wilson, G.G., Carlow, C.K.S., 1995. Molecular characterization of a cyclosporin A-insensitive cyclophilin from the parasitic nematode *Brugia malayi*. Biochemistry 34, 11545–11550.
- Perler, F.B., Comb, D.G., Jack, W.E., Moran, L.S., Qiang, B., Kucera, R.B., Benner, J., Slatko, B.E., Nwankwo, D.O., Hempstead, S.K., Carlow, C.K.S., Jannasch, H., 1992. Intervening sequences in an Archaea DNA polymerase gene. Proc. Natl. Acad. Sci. USA 89, 5577–5581.
- Perler, F.B., Davis, E.O., Dean, G.E., Gimble, F.S., Jack, W.E., Neff, N., Noren, C.J., Thorner, J., Belfort, M., 1994. Protein splicing elements: inteins and exteins a definition of terms and recommended nomenclature. Nucleic Acids Res. 22, 1125–1127.
- Roberts, G.A.F. and Taylor, K. (1988) The preparation and characterization of chitin beads for the use in chromatography. In: Skyjåk-Bæk, G., Anthonsen, T. and Sanford, P. (Eds.), Chitin and Chitosan Sources, Chemistry, Biochemistry, Physical Properties, and Applications. Elsevier Applied Science, London, pp. 577–583.
- Shao, Y., Xu, M.-Q., Paulus, H., 1995. Protein splicing: characterization of the aminosuccinimide residue at the carboxyl terminus of the excised intervening sequence. Biochemistry 34, 10844–10850.
- Shao, Y., Xu, M.-Q., Paulus, H., 1996. Protein splicing: evidence for an N-O acyl rearrangement as the initial step in the splicing process. Biochemistry 35, 3810–3815.
- Shumacher, T.N.M. and Tsomides, T.J. (1996) In vitro radiolabeling of peptides and proteins. In: Coligan, J.E., Dunn, B.M., Ploegh, H.L., Speicher, D.W. and Wingfield, P.T. (Eds.), Current Protocols in Protein Science, Vol. 1. John Wiley and Sons, New York, pp. 3.3.1–3.3.19.
- Smith, D.B., Johnson, K.S., 1988. Single-step purification of polypeptides expressed in *Escherichia coli* as fusions with glutathione S-transferase. Gene 67, 31–40.
- Stofko-Hahn, R.E., Carr, D.W., Scott, J.D., 1992. A single step purification for recombinant proteins. Characterization of a microtubule associated protein (MAP 2) fragment which associates with the type II cAMP-dependent protein kinase. FEBS Lett. 302, 274–278.
- Studier, F.W., Rosenberg, A.H., Dunn, J.J., Dubendorff, J.W., 1990. Use of T7 RNA polymerase to direct expression of cloned genes. Methods Enzymol. 185, 60–89.
- Van Dyke, M.W., Sirito, M., Sawadogo, M., 1992. Single-step purification of bacterially expressed polypeptides containing an oligohistidine domain. Gene 111, 99–104.
- Watanabe, T., Ito, Y., Yamada, T., Hashimoto, M., Sekine, S., Tanaka, H., 1994. The role of the C-terminal domain and type III domains of chitinase A1 from *Bacillus circulans* WL-12 in chitin degradation. J. Bacteriol. 176, 4465–4472.
- Wilkie, S.D., Landry, D., 1988. Determination of certain agarose immobilized ligands by the use of perchloric acid. BioChromatography 3, 205–214.
- Xu, M.-Q., Perler, F.B., 1996. The mechanism of protein splicing and its modulation by mutation. EMBO J. 15, 5146–5153.
- Xu, M.-Q., Southworth, M.W., Mersha, F.B., Hornstra, L.J., Perler, F.B., 1993. In vitro protein splicing of purified precursor and the identification of a branched intermediate. Cell 75, 1371–1377.
- Xu, M.-Q., Comb, D.G., Paulus, H., Noren, C.J., Shao, Y., Perler, F.B., 1994. Protein splicing: an analysis of the branched intermediate and its resolution by succinimide formation. EMBO J. 13, 5517–5522.